Archie P. H. Sinaga

Software Engineer | Next.js, Express.js, Supabase | Python
Palembang, ID | +62 812-5023-3529 | archiesinaga9@gmail.com | linkedin.com/in/archie-ph-sinaga | archiesinaga.github.io

EDUCATION

Universitas Sriwijaya **Indralaya, ID**
Bachelor of Computer Science (S.Kom.), Informatics, GPA 3.92/4.00 *June 2026*
- Awards: Third Best Presentation Team (Bitlabs, MSIB Batch 7); UI/UX Finalist (Technology Euphoria 2024).

WORK EXPERIENCE

Alpha Omega Learning Center (ALC) **Palembang, ID**
Data Analyst (part-time until June 2026, full-time since July 2026) *September 2025 – Present*
- Launched a Learning Management System tracking daily grades for 600+ students within 1 month (September 2026) by building it on Express.js and Supabase (PostgreSQL).
- Replaced manual QRIS payment checks for 150 students by integrating Xendit QR payments (JavaScript) into the LMS, so each student's monthly tuition is marked paid or unpaid automatically.
- Drove 120 new enrollments in November 2025 by tracking each new student's source on sign-up forms across social ads, referral, word of mouth, banners, and returning students; inbound Instagram inquiries converted at 50%+.
- Brought in 10 new students by designing a referral program that gave one free month of tuition for every 2 friends referred.

Tutor, Mathematics & Physics (Part-time) *July 2024 – May 2025, September 2025 – June 2026*
- Helped ~15 Grade 12 students reach universities including ITS, Universitas Sriwijaya, USU, and UNESA by teaching UTBK/SNBT Mathematics and Physics to ~60 students a week and tracking every student's scores in a structured tracker.

PT Sumatera Prima Fiberboard **Indralaya, ID**
IT Intern *June 2025 – August 2025*
- Cut document approval time from ~1 week to 2-3 days across 8 divisions by building a Next.js and Tailwind CSS approval system end to end with one partner, from backend design to interface and testing.
- Replaced estimated truck departure and stop times with GPS-recorded ones for 10-20 trucks a day across 4 checkpoints by building a trip-tracking web app (Next.js, TypeScript, MongoDB, Leaflet) with route maps, timed stops, and speeds.

Bitlabs Academy, MSIB Batch 7 **Online**
Data Analytics for Business Cohort *August 2024 – December 2024*
- Found that the most-visited customers generated 13x more revenue per customer by analyzing 6 months of a rural-commerce startup's sales, visit, and customer data with SQL, Python, Excel, and Tableau.

PROJECTS

CekSawit, Palm Oil Disease Expert System (Next.js, TypeScript, Tailwind CSS) | GitHub **November 2025**
- Designed a 25-rule knowledge base for 7 palm oil diseases from interviews with an agricultural expert (25+ years) and built, in a team of 5, a web app that scores 8 symptoms with certainty factors and returns a certainty % and treatment steps.

Bacteria Image Classifier, Undergraduate Thesis (Python, TensorFlow, ConvNeXt, ResNet-50) | GitHub **April 2026**
- Reached 99% accuracy and F1 on 2,033 microscope images of 33 bacteria species (DIBaS) by comparing ConvNeXt-Tiny with ResNet-50 (98.7%) across 6 training setups; deployed as a Streamlit app.

Credit Risk Model & Early Warning Dashboard (Python, XGBoost, SHAP, Streamlit) | GitHub **October 2026**
- Cut expected loss cost by 36.2% versus approving everyone by building a credit scoring model on 150,000 borrowers that reached 0.861 test AUC (XGBoost).
- Concentrated ~55% of defaults into the riskiest 10% of borrowers by designing A-E risk grades, shown in a Streamlit dashboard with SHAP-based risk reasons.

LEADERSHIP EXPERIENCE

Student Executive Board (BEM), Universitas Sriwijaya **Indralaya, ID**
Staff, Daily Executive Board *February 2024 – February 2025*
- Ran a faculty Mobile Legends tournament for 32 teams (170 players) and coordinated Rector's Cup selection for ~400 participants.

Isport Unsri (faculty sports club) **Indralaya, ID**
Head of Division *March 2023 – March 2024*
- Initiated the annual FASILKOM Games (futsal, chess, volleyball, basketball) for ~150 participants.

SKILLS

Languages & Frameworks: JavaScript, TypeScript, Python, SQL | Next.js, React, Express.js, Tailwind CSS, Streamlit
Databases & Tools: PostgreSQL (Supabase), MySQL, MongoDB | Git, Xendit API, Leaflet
Data & ML: pandas, scikit-learn, XGBoost, TensorFlow | Tableau, Power BI, Excel
Certifications: Data Analysis (SQL, Tableau, Power BI & Excel), Udemy | Data Analytics for Business, Bitlabs
Languages: Indonesian (native), English (Duolingo English Test 125)